🚀 PHASE 1 (4th–5th Sem)

🧠 1. CORE MACHINE LEARNING (Keep but focused)

Must Learn:

Supervised Learning

Unsupervised Learning

Feature Engineering

Model Evaluation

Algorithms (Don't go too deep, just practical):

Linear Regression

Logistic Regression

Decision Tree

Random Forest

KNN

SVM

K-Means

📚 2. LIBRARIES (Corrected Order)

1. NumPy
2. Pandas
3. Matplotlib
4. Seaborn
5. Scikit-learn

👉 Learn with hands-on mini tasks, not just syntax

👁️ 3. ADD-ON (VERY IMPORTANT 🔥)

This is where your plan becomes powerful.

Computer Vision

OpenCV

Image processing

Edge detection

Contours

Camera input handling

Basic Backend

FastAPI

REST APIs (basic)

👉 Don’t go deep, just enough to serve models later---

CSE Core (Light touch)

SQL basics

Basic NoSQL idea

REST concept

🧰 4. TOOLS

Jupyter Notebook

PostgreSql/MongoDB

🚀 5. 🔥 PROJECT SECTION (MINIMUM 10 – UPDATED)

🟢 Beginner (1–4)

1. Face Detection System

👉 Detect faces using webcam

2. Motion Detection System

👉 Detect movement (CCTV style)

3. Image Processing Toolkit

👉 Blur, edge, grayscale, filters

4. Hand Detection / Gesture Basic

👉 Detect hand using OpenCV

🟡 Intermediate (5–8)

5. Face Mask Detection

👉 Real-time classification

6. Object Detection (Basic OpenCV)

👉 Detect shapes / simple objects

7. Image Classifier (ML-based)

👉 Use sklearn (not deep learning yet)

8. Camera-based Intrusion Alert System 🔥

👉 Detect motion + trigger alert

🔴 Strong Projects (9–10)

9. Smart Surveillance Mini System 🔥

👉 Combine:

Motion detection

Human detection (basic)

Alert logic

10. ML Model API Project

👉 Input image → output prediction using FastAPI

⚠️ NOTE:

👉 These 10 = Phase 1 only foundation projects

👉 Later you will build advanced versions

🎯 6. INTERNSHIP SECTION (UPDATED & STRATEGIC)

🔥 Target Roles (Apply in Summer)

1. AI / ML Intern (Top Priority)

Computer Vision preferred

2. Data Analyst Intern (Backup)

Only if ML roles not available

3. Software Intern (Selective)

Only if AI-related backend

📍 Platforms to Apply

Internshala

LinkedIn

AICTE portal

🧠 Hidden Strategy (VERY POWERFUL)

👉 Email professors from:

IIT Bhubaneswar

NIT Rourkela

Ask for:

Research internship

AI / Vision project collaboration

🎯 Internship Readiness Checklist

Before applying you MUST have:

✅ 3–4 strong projects
✅ GitHub profile
✅ Basic ML understanding
✅ Resume (1 page)

⚡ FINAL REFINED PHASE 1 SUMMARY

You will have:

ML fundamentals ✓
Computer Vision start ✓

10 practical projects ✓

Internship-ready profile ✓